Table 13.3: Analysis of Variance for the One-Way ANOVA

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**	**Computed f**
Treatments	SSA	$k-1$	$s_1^2 = \dfrac{SSA}{k-1}$	$\dfrac{s_1^2}{s^2}$
Error	SSE	$k(n-1)$	$s^2 = \dfrac{SSE}{k(n-1)}$	
Total	SST	$kn-1$		

Three Important Measures of Variability

$$
\begin{aligned}
SST &= \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{..})^2 = \text{total sum of squares},\\
SSA &= n\sum_{i=1}^{k}(\bar{y}_{i.} - \bar{y}_{..})^2 = \text{treatment sum of squares},\\
SSE &= \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})^2 = \text{error sum of squares}.
\end{aligned}
$$

The sum-of-squares identity can then be represented symbolically by the equation

$$SST = SSA + SSE.$$

Problem 16.47

1) Ho : all four population mean are equal
2) H1 : two population mean are not equal
3) alfa = 0.05
4) Critical value and critical region

F critical = F 0.05, v1, v2 = F 0.05, 3, 8 = 4.07

$v1 = k - 1 = 4 - 1 = 3$

5) Test statistics (F- Test)

$v2 = k (n - 1) = 4(3 - 1) = 4(2) = 8$

	s1	s2	s3	s4	
	11	9	16	5	
	6	2	10	1	
	7	4	10	3	
Sum	24	15	36	9	$\sum x = 84$
Square	576	225	1296	81	$\sum (T^2) = 2178$

total obs = nt = 12
nj = 3
k= ni = 4

for SST Square each value

	121	81	256	25	
	36	4	100	1	
	49	16	100	9	
Sum	206	101	456	35	$\sum x^2 = 798$

$SST = \sum x^2 - (\sum x)^2/nt$

$SST = 798 - (84)^2/12 = 210$

for SSA $SSA = \sum (T^2/nj) - (\sum x)^2/nt$

$SSA = 2178/3 - (84)^2/12 = 138$

for SSE $SSE = SST - SSA$

$SSE = 210 - 138 = 72$

ANOVA table

		Sum Square	df	ms	F-ratio
	SSA	138	3	46	5.11
	SSE	72	8	9	
Total	SST	210	11		

Conclusion

F-ratio > critical value
5.11 > 4.07

Hence Ho reject